

Exam Date & Time: 03-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination, Jan-2025

Digital Computer Architecture [ECE-3124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Demonstrate the working principle of general register based computing machine using a neat block diagram showing the basic structure. List two example instructions for the above machine.
 - A) Implement 3- address instructions for computing the expression, $Y = (A - B) / (F * G - H) * ((C + D) / E)$ (assume memory operand is supported for source and destination). (5)
 - B) Demonstrate the working principle of direct, indirect, and relative addressing modes using one example for each. (3)
 - C) Write the short real IEEE format. Convert the decimal number 28.125 to 32-bit IEEE floating point representation. Write the final answer in hexadecimal. (2)
- 2) Apply Booth's Algorithm to perform the multiplication of -9 and -13, considering -9 as the multiplicand and -13.
 - A) (5)
 - B) Design a flowchart illustrating the steps of the unsigned binary division algorithm. Demonstrate a sample binary division problem and validate the result with the manual division. (3)
 - C) Explain the working mechanism of a Carry Look-Ahead Adder (CLA) and how it improves upon traditional ripple carry adders in terms of speed. (2)
- 3) Consider a 2-way set associative mapped cache of size 16 KB with block size 256 bytes. The size of the main memory is 128 KB. Calculate the size of the tag, set, and word.
 - A) (5)
 - B) Demonstrate the associative mapping mechanism of cache memory with a detailed example. Include a clear, labeled figure to illustrate the process. Also, discuss the need for cache memory in computer systems. (3)
 - C) Classify the different levels of the memory hierarchy in computer systems based on its speed, size, and cost. (2)

- 4) Demonstrate the concept of instruction-level parallelism (ILP) and how it improves performance. Using ILP 3 and ILP 3/2 as examples, demonstrate how instructions can be executed in parallel. Provide a scenario where this parallelism can be applied and explain its impact on execution efficiency. (5)
- A)
- B) Explain different types of bus structures within a CPU (single-bus, two-bus, and three-bus oriented ALUs). Provide an example for each to illustrate how data is transferred and processed. (3)
- C) Discuss the concept of data dependency hazards in instruction pipelines. Discuss different types of hazards (Data hazard, control hazard, instruction hazard). (2)
- 5) In a DSP system, you are using an 8-point Fast Fourier Transform (FFT). The memory addresses for your data range from 0 to 7. Using bit-reversed addressing, show the new order of these addresses.
- A) (5)
- a) What is bit-reversed addressing, and why is it used in FFT?
- b) Given the addresses 0 to 7, list the corresponding bit-reversed addresses for each one.
- B) Mention briefly about the fixed-point and floating-point processors.
- a) Find the key difference between fixed-point and floating-point processors in terms of data representation and processing capability.
- b) Provide an example of an application where each type of processor (fixed-point and floating-point) is more suitable. Explain why that particular processor is preferred in each application. (3)
- C) Explain why frequency domain analysis is performed in signal processing and how it differs from time domain analysis. (2)

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